TECHNISCHE UNIVERSITÄT MÜNCHEN

Quantum Chemistry Laboratory

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Full Solution of Exercise 8

	C-O distance Å	C-H distance Å	<H-C-O degrees	a_1 vibrational frequencies cm^{-1}	b_1 vibrational frequencies cm^{-1}	b_2 vibrational frequencies cm^{-1}
HF/STO-3G	1.217	1.101	122.8	1767 2099 3499	1279	1397 3646
HF/6-31G(d)	1.184	1.092	122.2	1680 2028 3160	1336	1383 3232
MP2/6-31G(d)	1.221	1.104	122.2	1583 1788 3013	1212	1296 3084

CISD/6-31G(d)	1.208	1.102	122.1	1605 1883 3039	1236	1318 3108
LSDA/6-31G(d)	1.206	1.126	122.4	1501 1838 2800	1146	1232 2841
B3LYP/6-31G(d)	1.207	1.111	122.4	1563 1848 2913	1198	1280 2965
Experimental Values	1.208	1.116	121.8	1563 1764 2944	1191	1288 3009

Differences up to 4 cm^{-1} between the listed frequencies and your calculated result are due to the not very tight stop criterion for geometry optimization and are normal. Small deviations in the geometry parameters too.

Aanswers to the Questions:

a. Becke3LYP

b. HF provides
   - shorter bond lengths
   - larger vibrational frequencies
   - a higher total energy in comparison to methods including electron correlation

Since the electons are more crowded the electron-electron repulsion energy is higher. This is the reason for the higher total energy.

The electrons are closer together, especially along chemical bonds. Because of the higher electron density the attractive forces between nuclei and electron cloud are enhanced and bond lengths shorten. Tighter bonds imply larger vibrational frequencies.

c. The small basis set STO-3G is very inflexible. Thus, the increased electron density between nuclei in molecules cannot be described in full extent, i.e. the calculated bond is somewhat longer than it should be. This effect and the contraction of the bond length due to higher electron density cancel each other out.

d. CPU times:

HF/STO-3G < HF/6-31G(d) < MP2/6-31G(d) < CISD/6-31G(d)

Inequation 1: STO-3G is the smaller basis set.

Inequation 2: MP2 includes a HF calculation. It is followed by a perturbation treatment.

Inequation 3:

- CISD consists of more determinants than MP2.
- CISD is iterative, MP2 non-iterative.
- With Gaussian the second derivative of the CISD wave function cannot be calculated analytically as in the case of HF and MP2. It must be calculated numerically which is more time consuming.

LSDA uses a rather rough approximation to describe the exchange-correlation functional, in which the electrons and nuclei are taken as homogeneously and uniformly distributed charges in space. Becke3LYP comprises spin-density gradients as well, so more parameters have to be calculated.

[General help] [To overview] [Compute] [Contact the course supervisor]

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